Progen ERP — Unified Functionality Specification

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Progen ERP is one platform: a shared core plus Construction & Builders, E-commerce and Organisations packs, with role-based access throughout. It is designed to close the gaps users report in Vyapar, Tally, Zoho, Odoo and the rest. For every feature, this spec says why it helps, who may use it and how to verify that it works, and it estimates what the build costs.

Market review

No established product combines Indian compliance, industry depth and a usable mobile app. Each covers one area well and leaves gaps at the edges. Vyapar and myBillBook bill fast but stop at one shop. Tally is the accounting standard but is tied to the desktop. Zoho, Odoo and ERPNext cover many modules but need partners, connectors or developers for Indian construction and marketplaces. Progen's opening is the combination.

Product	Best at	Price headline (ex-GST)	Main gaps users report
Vyapar	Fast GST billing for small shops; offline; cheap	Free mobile; desktop about ₹3,399/yr (reseller figure, unverified)	Fixed roles, no custom permissions; live sync on one device only; iOS behind Android; no payroll (reviews)
TallyPrime	Accounting depth, keyboard speed, CA familiarity	₹22,500 Silver / ₹67,500 Gold one-time, plus ₹4,500–13,500/yr support	Desktop-bound; mobile is view-only reports; slow on networks; dated UI
BUSY	GST billing and inventory for traders	₹11,000–22,000 one-time	Branch sync needs a static IP and SQL server, Enterprise only; licence drops to demo mode (reviews)

Product	Best at	Price headline (ex-GST)	Main gaps users report
myBillBook	Low-cost mobile billing	₹399/yr to ₹417/month	Data locked after the subscription lapses; reverse-charge GST errors; sync stuck (reviews)
Marg ERP	Pharma and FMCG distribution	₹5,550–26,000 plus yearly maintenance	Windows only; dealer-dependent support; multi-branch only in Gold (reviews)
Zoho Books / Zoho One	Cloud apps, GST, integrations	Books ₹749–7,999 per org/month; One ₹1,250–3,500 per user/month (yearly)	Apps joined by connectors (CRM to Books syncs every 2 hours); invoice caps; no native Flipkart; construction needs custom builds; online only
Zoho ERP (India, launched Jan 2026)	Single data model with payroll and GST	₹999 per user/month, minimum 5 users	New; not part of Zoho One
Odoo 20	One database for all modules; most detailed permissions	₹580–1,080 per user/month (first-year discount)	Upgrade every 2 years; custom modules must be ported; multi-company and API only on the higher plan; partner dependence; Amazon connector omits amazon.in
ERPNext v16	Open source, no licence fee	Free; hosting from about ₹410/month	Native mobile apps dropped in 2021; no offline POS; upgrades break customisations; no official construction app

Product	Best at	Price headline (ex-GST)	Main gaps users report
Procore / Buildertrend	Global construction project management	Priced by construction volume	No Indian payroll, GST, RERA or BOCW; offline daily logs stuck pending; complex for field staff
Powerplay / Onsite	Indian site apps: labour and materials in regional languages	Not published	Too many steps for field staff (reviews); no offline mode advertised; accounting through Tally sync
Sell.Do	Real-estate CRM, bookings, broker app	Not published	Support and recurring bugs; notes can't be updated on mobile (reviews)
Unicommerce / EasyEcom	Multichannel orders and warehouse	Quote only; API and reconciliation in top tier or as add-ons	Opaque pricing; slow support; long onboarding (reviews)
Keka / greytHR / Zoho Payroll / Darwinbox / factoHR	HR, payroll and attendance with Indian statutory rules	greytHR from ₹2,495/month for 50 employees; Keka and Darwinbox quote only	Geo-punches registered 100–200 m off site; night shifts marked absent; approval alerts arriving late (reviews); GPS tracking and face attendance sold as add-ons; HR only, no accounts or site costing

Gaps that recur across products

1. **Permissions an admin can configure without a developer**, by site, branch, field and amount.
2. **Multi-branch and multi-site in the base product**, not an enterprise upgrade.
3. **Offline mobile that syncs reliably** for sites, godowns and counters.
4. **The same features on mobile and desktop**, including data entry, not view-only reports.

5. **Indian construction built in:** BOQ, measurement book, RA bills, retention, labour, RERA.
6. **Native Indian marketplace connectors with settlement reconciliation.**
7. **Customisations that survive upgrades.**
8. **Customer owns the data:** full export at any time, even after a subscription lapses.
9. **Transparent flat pricing** with no invoice caps or top-tier-only APIs.
10. **Direct vendor support with service-level commitments**, instead of dealer dependence.

G2, Capterra and Reddit blocked automated access, so user evidence comes from app-store reviews, Trustpilot, PeerSpot and vendor forums. Prices are from vendor pages as of October 2026 and change often.

Product vision and platform architecture

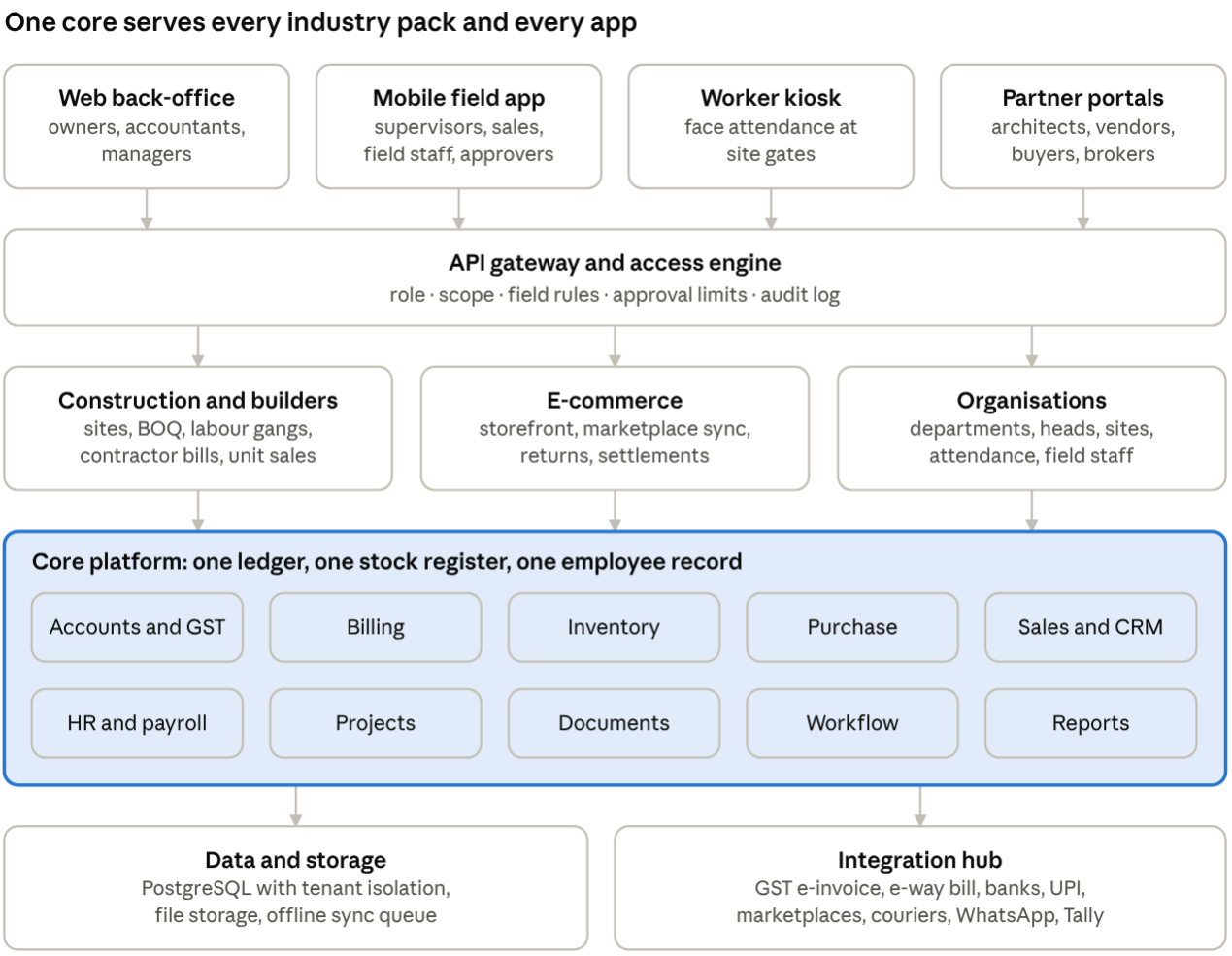
Progen is one ERP: a shared core (accounts, inventory, people, documents, workflow) plus industry packs you switch on per company. A builder who also sells online runs both from one ledger, one stock register and one login.

Design principles

1. **One core, many packs.** The Construction & Real Estate, E-commerce and Organisation packs add their own masters, screens, roles and reports. Every pack posts to the same ledger, stock and HR records, so nothing is re-entered.
2. **Hierarchy built in from day one.** Group → Company (one PAN / legal entity) → GST registration (one per state) → Branch → Site / Warehouse / Store / Department → sub-units (tower, floor, flat; rack, bin; team). Moving from 1 branch to 50 needs no migration.
3. **Every transaction carries dimensions.** Company, branch, site, project, cost centre, department, sales channel and salesperson are tagged on every voucher. Profit and loss by site, by channel or by department then comes from the ledger directly, not from spreadsheets.
4. **Configure, don't code.** Custom fields, form layouts, workflow and approval builder, print-template designer and report builder are available to an admin without a developer.
5. **Offline first where work happens.** Site, warehouse and billing-counter screens work without internet and sync when back online.
6. **API first.** Every screen uses the same public API. Every document emits events (created, approved, paid) that drive notifications, workflows and integrations.
7. **Built for India, ready for abroad.** GST, e-invoice, e-way bill, TDS/TCS, PF/ESI and Indian languages first; multi-currency and tax engines pluggable for export customers.

Platform layers

Layer	What it holds	Recommended technology
Client apps	Web back-office, Android/iOS field and approver apps, Windows billing desktop	Flutter (this repository) with role-based "modes": Admin, Supervisor, Worker kiosk, Sales, Delivery, Approver
Offline store	Local copy of the user's scoped data and an outbox of pending changes	SQLite via drift on device; outbox queue with server-side conflict rules
API gateway	Auth, tenant routing, rate limits, API keys for partners	REST + webhooks; OAuth2 / JWT; per-tenant API keys
Core services	Accounts, inventory, purchase, sales, CRM, HR & payroll, projects, documents, workflow, notifications	Modular monolith first (one deployable, strict module boundaries); split hot modules later
Industry packs	Construction & Real Estate, E-commerce, Organisation	Same codebase, enabled per company by licence flag
Integration hub	GSTN / IRP, e-way bill, banks, payment gateways, marketplaces, couriers, WhatsApp, Tally	Background workers + job queue with retries and a visible sync log
Data	Transactions, files, search, analytics	PostgreSQL with row-level security per tenant; object storage for photos and documents; read replica for reports



Progen architecture · 4 apps, 3 packs, 10 core modules

Every request passes the access engine before it reaches a pack or a core module. All three packs write to the same core, so one report can span construction, online sales and departments.

Role-based access control

Access is decided by four things together: **role** (which actions), **scope** (which companies, sites, departments, warehouses or channels), **field rules** (which fields are hidden, read-only or editable) and **limits** (how much they can approve or discount). One user can hold several role + scope pairs, for example Site Engineer on Site A and Store Keeper on Site B.

Permission layers

Layer	Options	Example
Action	View, create, edit, delete, cancel, approve, print, export, share, backdate, reopen	Accountant can cancel invoices; Sales Executive cannot
Record scope	Own, my team, my department, my site, my branch, my company, all	Site Supervisor sees only purchase indents of their own site
Field rule	Hidden, read-only, editable — per field, per role	Purchase rate and margin hidden from supervisors; salary hidden from everyone except HR and the employee
Limit	Amount, quantity, discount %, credit days	Project Manager approves purchase orders up to ₹2,00,000; above that goes to Director
Time	Start and end date on any assignment	Auditor access for 30 days; architect access ends when the project closes
Device and network	OTP / 2FA, device binding, IP allow-list, login hours	Labour kiosk works only on the registered site tablet

Rules the engine must enforce

1. **Deny by default.** A new role sees nothing until permissions are granted.
2. **Scope filters every list, report, search, export and API call.** A supervisor's "stock report" shows only their site, without a separate report.
3. **Approval matrix.** Multi-level, sequential or parallel, by amount and document type. Delegation during leave, escalation after a set number of hours, approve or reject from mobile or WhatsApp with a reason.
4. **Maker–checker.** The person who creates a payment, rate change or vendor bank-detail change cannot approve it.
5. **Period lock.** No entries before a locked date (for example after GST return filing) without an admin unlock that is logged.
6. **Complete audit trail.** Who changed which field, old value, new value, time, device and location; never deletable.
7. **"View as" for admins.** An admin can preview exactly what a role sees before assigning it.

8. **External users are first-class.** Architects, contractors, vendors, buyers, brokers and the CA log in to portals with the same engine, limited to their own records.

Default role templates

Every template is a starting point that the admin can copy and edit.

Role	Pack	Typically can	Typically cannot
Owner / Super Admin	All	Everything, including billing of the subscription and user management	—
Company Admin	All	Users, roles, masters, settings for one company	See other group companies
Accountant	Core	Vouchers, GST returns, bank reconciliation, reports	Change roles or delete audit history
CA / Auditor (external)	Core	Read-only books, reports, export	Create or edit any entry
Purchase Manager	Core	Vendors, RFQs, purchase orders up to limit	Approve own purchase orders
Store Keeper	Core	Goods receipt, issue, transfer, stock count for assigned stores	See purchase rates if hidden
Sales Executive	Core	Leads, quotes, sales orders, own customers	Give discount above limit, see cost price
Project Director	Construction	All projects and sites, budgets, approvals above manager limit	—
Project Manager	Construction	Assigned projects: budget, schedule, contractors, approvals up to limit	Other projects
Site Engineer / Supervisor	Construction	Daily progress report, labour attendance, material indent, issue, photos for own site	Rates, payments, other sites

Role	Pack	Typically can	Typically cannot
Quantity Surveyor / Billing Engineer	Construction	BOQ, measurement book, contractor and client running bills	Approve payments
Architect / Consultant (external)	Construction	Upload drawings, answer RFIs, approve work stages on assigned projects	Costs, payments
Contractor / Subcontractor (external)	Construction	View work orders, submit bills and labour lists, see payment status	Other contractors' data
Labour Contractor / Foreman	Construction	Mark attendance of own gang on kiosk or app	Wages of other gangs
Real-estate Sales Executive	Construction	Leads, site visits, unit holds and bookings	Change unit price list
Channel Partner / Broker (external)	Construction	Register leads, see own commission	Other brokers' leads
Buyer (portal)	Construction	Own unit, payment schedule, receipts, construction photos	Anything else
Catalog Manager	E-commerce	Products, listings, prices per channel	Settlements, refunds
Order / Fulfilment Manager	E-commerce	Orders, pick-pack-ship, courier allocation	Change prices
Picker / Packer	E-commerce	Scan-based picking and packing on the warehouse app	Any financial data
Returns Executive	E-commerce	Return receipt, quality check, restock or write-off	Issue refunds above limit
Marketplace Reconciliation	E-commerce	Settlement import, fee and claim matching	Change orders
Director / CEO	Organisation	Cross-department dashboards and final approvals	—

Role	Pack	Typically can	Typically cannot
Department Head (HOD)	Organisation	Own department's people, budget, requests, approvals	Other departments' salaries
Team Lead / Manager	Organisation	Team attendance, leave, tasks, expense approval up to limit	Payroll
Employee	Organisation	Self-service: attendance, leave, payslips, expenses, requests	Others' data
Field Employee	Organisation	Geo-tagged check-in, visits, orders, expenses from mobile	Back-office screens
HR Manager / Payroll Admin	Organisation	Employees, payroll, statutory filings	Accounts outside payroll

Core modules (every business)

Twelve core modules ship to every customer regardless of industry. The industry packs build on them rather than duplicating them.

1. Accounting and finance

Feature	Why it helps
Indian chart of accounts templates (trading, construction, e-commerce, services) with Schedule III financial statements	Ready on day one; a CA recognises the structure
All voucher types: sales, purchase, receipt, payment, contra, journal, debit and credit notes	Replaces Tally for daily bookkeeping
Bank feeds and auto-reconciliation that matches UPI, NEFT, IMPS and cheque references	Bank reconciliation in minutes instead of a day each month
Cheque register, post-dated cheques and bounce handling	Builders and distributors still collect large amounts by cheque

Feature	Why it helps
Cost centres and dimensions (site, project, department, channel) on every line	P&L per site, channel or department without spreadsheets
Budgets versus actuals with alerts at 80% and 100%	Stops cost overruns before they happen
Receivables and payables ageing with automatic WhatsApp and email reminders	Faster collection; fewer missed vendor dues
Credit limit and credit days per customer, enforced at order time	Prevents bad debt
TDS and TCS deduction at voucher level with challan and return data	No separate TDS software
Fixed assets with depreciation under Companies Act and Income Tax Act	Two depreciation books from one register
Multi-company consolidation with automatic inter-company entries	Groups with several firms see one combined picture
Cash-flow forecast from open orders, bills and payroll	Owner knows next month's cash gap in advance
Period lock and year-end closing with carry-forward	Books stay final after filing

2. Billing and invoicing

Feature	Why it helps
Counter-speed billing: barcode scan, item search, keyboard shortcuts, under 10 seconds per bill	Matches the speed shopkeepers like in Vyapar
Document chain: estimate → sales order → delivery challan → invoice → credit note, one click each step	No retyping; quantities tracked across the chain
E-invoice IRN and e-way bill generated from the invoice screen	No portal login, no double entry
Print designer: A4, A5, thermal 2″/3″, regional language, logo, terms, bank details, UPI QR	Every business has its own format

Feature	Why it helps
Share invoice on WhatsApp, email or SMS with payment link and UPI QR	Customers pay faster
Recurring invoices, proforma, export invoices in foreign currency	Covers rentals, AMCs and exporters
Price lists per customer group, schemes, slab discounts, free quantity	Distributors and wholesalers need these daily
Partial payments, advances and on-account adjustments	Real-world payment behaviour

3. Inventory and warehouse

Feature	Why it helps
Unlimited warehouses, sites and stores, each with racks and bins	Same system for a shop, a godown and a construction site
Variants (size, colour, grade) and unit conversion (bag ↔ kg, box ↔ piece, truck ↔ cubic metre)	Cement, steel, sand and apparel all buy in one unit and use in another
Batch, expiry, serial and IMEI tracking	Pharma, electronics and warranty items
Barcode and QR label printing, mobile scanning for receipt, issue and counting	Fewer counting errors
Reorder level, safety stock and auto-indent	Never run out of fast movers
Stock transfer with in-transit status and receiving confirmation	Stock between sites or warehouses never disappears
Valuation by FIFO or weighted average; landed cost (freight, loading, duty)	Correct margins and closing stock
Kits, bill of materials and simple assembly	Combo packs for e-commerce; fabricated items for construction
Physical stock audit on mobile with variance approval	Annual and surprise audits
Wastage, scrap and theft write-off with reasons	Real loss visible per site

4. Purchase and procurement

Feature	Why it helps
Indent → RFQ to several vendors → comparative statement → purchase order → goods receipt → quality check → bill	Full control from need to payment
Three-way match of PO, goods receipt and bill before payment	Pay only for what was ordered and received
Vendor portal: vendors quote, accept POs, upload invoices and see payment status	Fewer calls; faster cycle
Rate contracts and price history per item per vendor	Spot overpricing
Vendor rating on price, delivery time and quality rejection	Choose better suppliers
Advances against PO with automatic adjustment	Common for steel and cement

5. Sales and CRM

Feature	Why it helps
Lead capture from website, WhatsApp, Facebook/Instagram ads, IndiaMART, JustDial, 99acres, MagicBricks, Housing.com	All enquiries land in one list automatically
Pipeline board, follow-up reminders, call and visit logging with GPS	Nothing slips; managers see real activity
Customer 360: orders, invoices, payments, tickets, conversations in one screen	Anyone can answer a customer call
Territories, targets and commission rules per salesperson	Motivates and pays the sales team correctly
Service tickets, AMC contracts and warranty tracking	After-sales handled in the same system

6. HR and payroll

Employee master, attendance, leave, shifts and payroll with statutory deductions. Construction labour wages and department-level detail are in the industry packs; statutory rules are in Compliance.

7. Projects and tasks

Feature	Why it helps
Projects, tasks, sub-tasks, dependencies, Gantt and Kanban	Plan and follow up in one place
Timesheets linked to payroll and client billing	Service firms bill hours accurately
Billing by milestone, fixed price or time and material	Matches how contracts are written
Project P&L with committed costs (open POs) and actual costs	True profit while the project is running, not after

8. Documents, approvals and communication

Feature	Why it helps
Attach files and photos to any record, with versions	Drawings, bills and agreements next to the transaction
OCR: photograph a vendor bill and get a draft purchase entry	Saves typing; fewer errors
Aadhaar eSign and DSC signing of documents	Agreements and offer letters signed without printing
Comments and @mentions on any record; in-app chat	Discussion stays with the record, not lost in WhatsApp
Notifications by push, WhatsApp, SMS and email, configurable per role	Right person, right channel

9. Reports and analytics

Feature	Why it helps
Standard reports for every module, drill-down from total to voucher	Answers most questions without customisation
Drag-and-drop report and dashboard builder	Admins build their own views
Role-based home dashboards (owner, site, warehouse, HOD)	Each person sees what they act on
Scheduled reports by email or WhatsApp (daily cash, weekly site cost)	Owners stay informed without logging in
Ask in plain language ("material cost on Site B this month")	Owners who don't use reports still get answers

10. Assets and maintenance

Asset register with QR tags, assignment to employee or site, maintenance schedules, AMC and insurance renewals, and depreciation linked to accounts.

11. Expenses

Employees snap a receipt, choose a category and project; the claim goes through approval and is paid with payroll or by bank transfer. Petty cash per site with top-up requests.

12. Retail POS (optional)

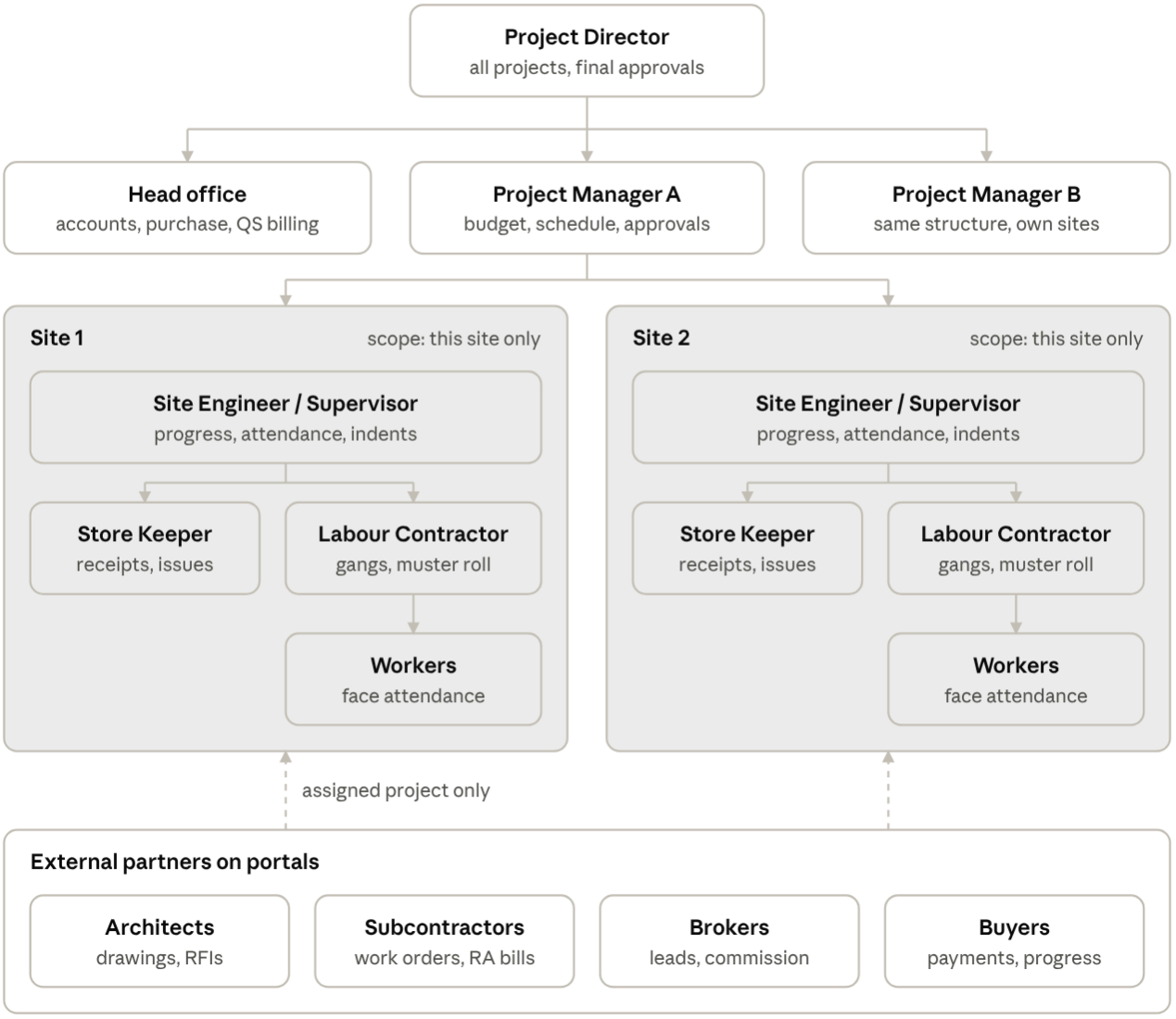
Touch billing, offline counter mode, cash drawer and shift closing, loyalty points and gift vouchers, synced with central stock.

Construction and builders pack

This pack runs every site from estimate to handover, and every flat from enquiry to possession, with labour, contractor, GST and RERA rules built in. Indian site apps such as Powerplay and Onsite cover site work but stop short of full accounting. Global tools such as Procore list no Indian wage payroll, GST, RERA or BOCW features.

Each site is a data boundary; partners see only their assigned project

Solid line: reports to. Dashed line: portal access, limited by role and scope.



Construction roles · director to workers, with partner portals

A supervisor on Site 1 never sees Site 2's stock, labour or costs. Head office sees every site, and partners log in only to the project they are assigned to.

Projects, sites and planning

Feature	Why it helps
Project → site → tower or block → floor → unit or activity structure; many sites per project, many projects per company	Matches how builders actually organise work

Feature	Why it helps
BOQ with rate analysis (material + labour + equipment + overhead per item), version history, Excel import	Estimate once; every later cost is compared against it
Budget by cost code showing committed (open POs and work orders), actual and forecast, alerts at 75%, 90% and 100%	Overruns are caught while they can still be fixed
Gantt schedule with baseline versus actual, a weekly look-ahead for supervisors, earned-value indicators (cost and schedule performance)	Shows early whether a site is slipping
Variation orders approved by the client, updating BOQ, budget and client billing together	Extra work always gets billed

Site execution (supervisor app, works offline)

Feature	Why it helps
Daily progress report: quantity done per BOQ item, labour count by trade, equipment hours, material used, weather, hindrances, geo-tagged and time-stamped photos; "copy yesterday" and voice notes	Owners see real progress every evening without visiting
Tasks and issues assigned to a contractor or worker, closed only with a photo	Clear accountability on site
Quality checklists per activity (for example, pre-pour slab check), test results such as cube tests, snag lists	Quality evidence for clients and RERA
Safety: toolbox talks, PPE checks, permits to work, incident reports	Fewer accidents; proof of compliance
Drawing register with revisions; site sees only the latest approved revision; RFIs to architects with due dates	No work done to an outdated drawing
Site petty cash with photo of every expense and top-up requests	Ends untraceable cash spend at sites

Labour and workforce

The new Labour Codes took effect on 21 November 2025. The principal employer is liable for unpaid contractor wages and jointly for PF and ESI, so Progen tracks contractor compliance, not only attendance.

Feature	Why it helps
Worker register: photo, Aadhaar (with consent), trade, skill, daily rate, labour contractor or gang, bank or UPI, BOCW registration number, safety training	One record per worker across all sites
Face attendance on a site kiosk or the supervisor's phone, with geo-fence and offline mode; bulk attendance by gang	Stops proxy attendance; works in basements and remote sites
Overtime with multipliers, half days, shifts	Wages match the real hours worked
Muster roll and wage, overtime and deduction registers produced per labour contractor	Statutory registers ready for inspection
Wages by day, piece rate or measured quantity; advances and deductions; bulk payment by bank or UPI; payslip by SMS or WhatsApp in the worker's language	Workers are paid correctly and on time
Alerts when a contractor has not paid wages by the due date, so the principal employer can pay and recover	Avoids legal liability for the builder
Contractor licence, PF and ESI challans checked before releasing a contractor's bill	The builder is not left holding the contractor's dues
Labour productivity: output per worker-day by trade, against norms	Shows which gangs and contractors perform

Materials, stores and equipment

Feature	Why it helps
Site indent → approval against remaining BOQ budget → central PO or local purchase → site goods receipt with photo of challan, vehicle and weighbridge slip	Stops over-ordering and fake deliveries

Feature	Why it helps
Issue to activity or contractor; theoretical versus actual consumption with wastage % per item	Material leakage becomes visible
Free-issue material to contractors with automatic recovery for excess wastage	Contractors pay for what they waste
Transfers between sites and stores with in-transit status	Surplus at one site supplies another
Returnable items (shuttering, scaffolding, tools) tracked by who holds them	Expensive reusable items stop disappearing
Equipment register, own and hired: hour meters, diesel log per machine, maintenance; hire bills generated from the logs	Diesel and hire charges are among the biggest leaks

Contractors and billing

Feature	Why it helps
Work orders to subcontractors with rates per BOQ item	Clear scope and price before work starts
Measurement book: site engineer records quantities, quantity surveyor checks them	Bills are based on verified quantities
Running-account (RA) bills: cumulative quantity, previously billed, this bill; retention held and released after the defect period; advance and material recovery; penalties	Contractor bills calculated correctly every time
Automatic TDS on contractor payments (under both the old section 194C and the new section 393 of the Income-tax Act 2025) and BOCW cess deduction where applicable	Statutory deductions never missed
Contractor portal: submit bills and labour lists, track approval and payment	Fewer follow-up calls; faster billing cycle
Client billing by RA, milestone or schedule of values with GST; certification by the client's engineer; retention receivable tracking	Builders and contractors get paid for every stage

Feature	Why it helps
Project P&L and cash flow: budget versus committed versus actual	True profit while the project is running

Real-estate sales (builders)

Feature	Why it helps
Unit inventory chart by tower and floor: carpet area (RERA basis), facing, parking; hold with expiry, block, book; launch-day booking with no double booking	Sales team never sells the same flat twice
Leads from property portals, website and ads, de-duplicated; site visits verified by OTP; cost sheet with floor rise, preferred-location charges, parking, GST and stamp-duty estimate	Faster, consistent quotes
Booking with KYC and co-applicants; warning if the advance exceeds the 10% limit before an agreement for sale; agreement generation; home-loan tracking	Stays within RERA from day one
Payment plans (construction-linked, time-linked, down payment); demand letters raised automatically when the site marks a stage complete; reminders, interest on delay, receipts	Collections follow construction without manual follow-up
RERA: 70% of collections routed to the separate project account; withdrawals matched to completion with engineer, architect and CA certificates; quarterly status update pack	Compliance reports come from live data
Broker portal: KYC, lead registration with ownership protection, commission slabs, TDS, payout only after the customer pays	Fair, transparent broker payouts
Buyer app: unit, payment schedule, receipts, documents, construction photos, change requests with cost, snag reporting at possession	Fewer calls to the sales office; happier buyers

Feature	Why it helps
Cancellation and refund with deductions as per the agreement; handover to the society and defect-liability complaints after possession	Covers the full life of the sale

E-commerce pack

Progen becomes the single record of catalogue, stock, orders, returns and money across the seller's own website and every marketplace. Amazon, Flipkart, eBay, Shopify and WooCommerce can be connected directly through public APIs. Meesho, Myntra, Ajio, Nykaa and JioMart only issue credentials to approved partners, seller by seller, so Progen needs a partner agreement with each and a guided onboarding flow.

Channels and what can be integrated

Channel	Integration route	What syncs	Limits to design around
Own website	Built into Progen	Everything	—
Shopify	Public GraphQL Admin API (REST is legacy; new public apps must use GraphQL)	Products, orders, inventory, fulfilment	Cost-based rate limits
WooCommerce	Public REST v3 + webhooks on the seller's site	Orders, refunds, products, variations, customers, coupons	Each site's hosting speed varies
Amazon	Public Selling Partner API; Progen registers as a public developer app	Orders, listings, catalogue, FBA stock, finances, reports, Easy Ship labels	Amazon approves data roles; developer fees were withdrawn in May 2026
Flipkart	Seller API; partner app needs Flipkart approval	Orders, labels, invoices, manifests, returns (incl. RTO), listings, price, stock per location	Settlement reports API is paid and partner-only — fall back to file import

Channel	Integration route	What syncs	Limits to design around
Meesho	Partner-only; credentials issued per seller (details)	Catalogue, stock (one facility), single-item orders, labels, returns	No invoice or tax amounts in the API; orders pushed by Meesho; settlements by file
Myntra (PPMP)	Partner-only via the seller's account manager (details)	Orders, Myntra-generated invoice and label, stock, returns	No catalogue pull; Myntra Logistics only
Ajio	Partner-only (POB ID) (details)	JIT orders, invoices, manifests	Stock moves by transfer to Ajio's warehouse; returns handled manually
Nykaa / Nykaa Fashion	Partner-only (details)	Catalogue import, orders	Full scope not public
JioMart	Partner-only (details)	Orders, stock by location, labels, returns	JioMart picks the fulfilling location; no order splitting
eBay	Public developer programme	Listings, orders, tracking, returns	eBay India closed in 2018, so this is export selling only: export invoice, LUT, foreign currency

Catalogue and pricing

Feature	Why it helps
One master catalogue: variants, images, attributes, channel-specific titles, prices and images; bulk edit by Excel	Edit once, publish everywhere
SKU mapping of ASIN, FSN, Myntra style ID and others to the master SKU; combos map to their components	Stock stays right even when channels name items differently

Feature	Why it helps
Fee calculator per marketplace: commission, fixed fee, shipping, collection fee, GST on fees	Seller sees real profit before setting a price
Floor-price guard: alert or block when a channel price falls below break-even	Stops loss-making listings
Listing health: suppressed, inactive or out-of-stock listings flagged daily	Recovers lost sales quickly

Orders and inventory

Feature	Why it helps
One order inbox for all channels with a countdown to each dispatch deadline	No late-dispatch penalties
Real-time stock sync with a safety buffer and allocation per channel (for example, keep 20% for the own website)	Prevents overselling and cancellations
Handles both pulled orders and orders pushed by the marketplace; every order processed exactly once	Meesho-style push and Amazon-style pull both work without duplicates
Order routing to the nearest warehouse with stock; FBA and other marketplace warehouses kept as separate virtual warehouses	Faster delivery; marketplace-held stock always visible
B2B, wholesale and dropship orders from the same inbox	One system for all sales

Warehouse: pick, pack, ship

Feature	Why it helps
Wave and batch picking on mobile; scan to pick and scan to pack, blocking a wrong item	Fewer wrong shipments
Pack-station video and weight photo saved against each AWB number	Evidence for marketplace claims and courier weight disputes

Feature	Why it helps
Invoice, label and manifest in one click; channel-generated labels where the channel requires it	Packing line keeps moving
Courier allocation for own-site orders by pincode, weight, COD and cost, with NDR (failed delivery) follow-up	Cheaper, more reliable shipping

Returns and RTO

COD orders return to origin (RTO) far more often than prepaid ones: about 26% against under 2%, per Shipway data cited by Amazon. Progen treats RTO as a cost to manage, not an afterthought.

Feature	Why it helps
Customer returns and RTO tracked separately; returns received by scan	Different causes, different fixes
Quality grading (good, damaged, wrong item, missing part) → restock, refurbish or write off	Returned stock is back on sale the same day
Claim tracker with evidence attached (pack video, photos) for Amazon SAFE-T, Flipkart and courier claims	Recovers money sellers usually give up
COD risk score from pincode, customer history and order value; WhatsApp confirmation or prepaid nudge for risky orders	Fewer RTOs before they ship
RTO and return dashboard by channel, courier, pincode and SKU	Shows exactly where losses come from

Settlements and accounting

Feature	Why it helps
Settlement import (API or file) for each channel, matched line by line: order → invoice → commission, fees, ads, TCS, TDS → payout	Finds underpayments automatically; replaces hours of monthly spreadsheet work

Feature	Why it helps
Automatic accounting: gross sales, marketplace fees with GST input credit, TCS and TDS receivable, returns in the right period	Books are correct without an accountant re-entering data
TCS matching with the marketplace's GSTR-8 and a reminder to accept the credit on the GST portal	Unclaimed TCS credit is recovered
Sales booked against the right state GSTIN, including marketplace warehouses in other states	Avoids GST mismatch notices
Channel and SKU profitability after fees, returns, RTO and ads	Shows which products and channels actually make money

Own website and social selling

Feature	Why it helps
Built-in storefront on the seller's domain: themes, catalogue from the master, cart, coupons, reviews, SEO pages	A brand website without a separate platform fee
Checkout with UPI, cards, wallets and COD through Razorpay, Cashfree, PhonePe or PayU	Familiar payment options for Indian buyers
Abandoned-cart and order-update messages on WhatsApp	Recovers sales; fewer "where is my order" calls
B2B portal: customer-specific prices, credit limits, bulk ordering	Serves dealers and wholesalers online
WhatsApp catalogue and Instagram shop fed from the master catalogue	Social sales without separate stock lists

Organisations pack

The Organisations pack runs people, departments and sites for any company with more than one department or location: offices, factories, schools, hospitals, agencies and service firms. Every feature respects the role scopes above, so a department head manages only their department.

Structure and departments

Feature	Why it helps
Org builder: company → branch → site → department → sub-department → team, with designations and grades	Mirrors the real organisation; drives permissions and reports
Department heads, deputies and acting heads during leave	Approvals never get stuck
Matrix reporting: an administrative manager and a functional manager per employee	Site staff report to the site head and to their function head
Live org chart with headcount, vacancies and cost per box	Leaders see gaps at a glance
Department budget, headcount plan versus actual, department cost and P&L	Department heads become accountable for spend
Department inbox for requests (IT, admin, HR, purchase) with SLA	Requests stop getting lost on WhatsApp

Workforce across sites

Feature	Why it helps
Assign employees to sites with transfer and deputation history	Know who is where, and since when
Multi-site shift rostering with drag-and-drop and conflict warnings	Avoid double booking and understaffed shifts
Agency and contract staff tracked separately, with agency invoice matching to attendance	Pay agencies only for days actually worked
Site-wise headcount, attendance and labour cost dashboards	Compare sites fairly

Attendance and field staff

Feature	Why it helps
Biometric device integration (ZKTeco, eSSL and similar)	Uses hardware companies already own

Feature	Why it helps
Face attendance on phone or shared kiosk tablet with liveness check	Stops buddy punching without hardware cost
Geo-fence per site and office; QR code at the gate	Punch only when physically present
Offline punch stored on device and synced later	Works in basements and remote sites
Overtime, comp-off, late-coming and short-leave rules per policy	Payroll gets clean data
Regularisation requests with manager approval	Forgotten punches fixed properly
Field staff: route plan, visit check-in with photo and GPS, orders and collections from the field	Managers see real visits, not claims
Location tracking only during duty hours, with consent	Respects employee privacy and the DPDP Act
Automatic distance calculation for travel reimbursement	Ends disputes over kilometres claimed
Geo-fence accuracy check: radius per site, minimum GPS accuracy, several sites per employee, Wi-Fi fallback indoors	Competitor apps accept punches 100–200 m away or reject staff standing inside the office
Night shifts that cross midnight counted as one shift	Night-shift staff are not wrongly marked absent
Battery-efficient tracking that keeps working when the phone puts the app to sleep, with accurate distance	Field-tracking apps drain batteries and under-count kilometres
Missed-punch reminder and punch-status widget on the phone; tracking status always visible to the employee	Fewer regularisation requests; tracking feels transparent, not like surveillance

Leave, payroll and self-service

Feature	Why it helps
Leave policies per location, grade and employment type; state-wise holiday calendars	One system for staff in many states
Payroll for monthly, daily-wage, hourly and piece-rate workers	Covers office staff and factory or site workers alike
Salary advances and loans with EMI deduction	Common need; no side registers
Employee app: payslips, leave, attendance, expenses, tax declarations, Form 16, requests	HR answers fewer routine questions
Approvals hub: one inbox for leave, expense, purchase, travel and payment approvals	Managers approve everything from one screen

Hiring to exit

Feature	Why it helps
Manpower requisition by HOD → approval → job post → candidate pipeline → offer	Hiring follows budget and approval
Digital onboarding: documents (Aadhaar, PAN, bank) collected on mobile, asset issue, induction checklist	New joiners productive on day one
Goals and OKRs cascading company → department → person; 360 reviews; appraisal linked to increments	Fair, documented performance decisions
Training calendar and certification tracking with expiry alerts (safety, licences)	Mandatory training never lapses
Exit: resignation, notice period, clearance by each department, full and final settlement, letters	Clean exits with nothing pending

Workplace and communication

Feature	Why it helps
Announcements by department or site; policy acknowledgement tracking	Proof that staff read the policy
Visitor management: pre-registration, gate pass, host notification	Security and a record of every visitor
Meeting-room and vehicle booking	Shared resources without clashes
Pulse surveys and anonymous feedback	Early warning on morale
People analytics: headcount, attrition, absenteeism, overtime and cost per employee by department and site	HR and leaders act on facts

Improvements competitors don't provide well

These are the reasons a customer switches. Each closes one of the gaps in the market review, or removes a daily pain that users currently handle in WhatsApp, Excel or a second app.

Improvement	What it does	Why users need it
One login for mixed businesses	A builder who also runs a hardware shop and an online store uses one company group, one ledger, one stock register	Indian business families rarely run one line of business; today they buy three tools
No-code permission engine with "view as"	Admin sets site, field and amount rules on one screen and previews exactly what a role sees	Vyapar's roles are fixed; Odoo needs developer mode; Zoho sets roles app by app
Truly offline mobile	Every field screen works offline for days; per-device number blocks; conflicts go to a review queue	Sites, basements and godowns have weak signal; competitors' offline sync gets stuck

Improvement	What it does	Why users need it
WhatsApp as a working channel	Approve a PO, get the daily site report, send invoices, collect payment, mark attendance or raise an indent by WhatsApp	Owners and supervisors already live in WhatsApp; approvals stop waiting for a laptop
Voice entry in Indian languages	A supervisor speaks the daily report in Hindi or Tamil; Progen fills the structured form for review	Many supervisors are slow at typing; reports actually get filed
Fraud and leakage alerts	Flags the same face on two sites on one day, duplicate vendor bills, rates above contract, diesel use out of line with machine hours, stock issued beyond BOQ	Ghost workers, diesel theft and duplicate bills are top leakages on sites
Photo-to-entry	Vendor bill, delivery challan or weighbridge slip photographed and turned into a draft entry	Removes most manual typing at site and in accounts
Live cost per BOQ item	Every PO, goods receipt, labour day and contractor bill updates committed and actual cost against the BOQ line	Owners know profit per project now, not at year-end
Proof-linked buyer demands	A demand letter goes to the buyer with site photos of the completed stage attached	Buyers pay faster when they see progress; fewer disputes
True profit per SKU and channel	Settlement reconciliation, RTO cost, ads and fees included in the base e-commerce pack	Aggregators sell reconciliation as an add-on; sellers don't know which SKUs lose money
Accountant workspace	The CA gets one login across all their client companies, a review queue, comments on vouchers and GST return views	Small businesses depend on their CA; Tally-style file sharing ends

Improvement	What it does	Why users need it
Compliance calendar	Due dates for GST, TDS, PF, ESI, professional tax, RERA updates and licence renewals, with reminders and penalty estimates	Missed deadlines cost late fees and interest
Upgrade-safe customisation	Custom fields, forms, workflows and reports stored as configuration and checked on every release	Odoo and ERPNext customers pay to re-port customisations at each upgrade
Go live in a day	Importers for Tally, Vyapar, BUSY and Excel, plus a parallel-run mode that compares Progen's books with the old system	Fear of migration keeps businesses on old tools
Data ownership guarantee	Read-only access and full export stay available after a subscription ends	myBillBook and BUSY lock data behind renewal
Transparent pricing	Price per company plus optional packs; no invoice caps; API and portals included	Competitors cap invoices, gate APIs to top tiers or quote only
Practice mode	A sandbox copy with demo data for training new staff	New staff learn without touching real books
Worker-friendly kiosk	Icon-only screens with voice prompts in the local language	Usable by workers with limited literacy

Compliance and integrations

Indian law changed heavily in 2025–26: GST rates were rationalised, the Labour Codes and the Income-tax Act 2025 took effect, and the DPDP Rules were notified. Progen must store every rule with an effective date, so old documents keep old rules and new documents follow new ones. Items marked "verify" rest on secondary sources and need a check against the official text before build.

Compliance rules to build (as of October 2026)

Area	Rule	What Progen must do
GST rates	5% and 18% slabs plus a 40% rate from 22 Sep 2025	HSN rate master with effective dates; old invoices keep old rates
E-invoice	Mandatory above ₹5 crore turnover (verify); IRN must be obtained within 30 days for turnover of ₹10 crore and above	IRN and signed QR from the invoice screen; warning before day 30
E-way bill	Multi-factor login; only for documents up to 180 days old; extension capped at 360 days. Ship-to GSTIN and closure changes announced, then put on hold	Generate and extend from invoice or transfer; build the on-hold changes behind a switch
Invoice Management System	Accept, reject or keep pending each supplier invoice; no action counts as accepted when GSTR-2B is generated	IMS inbox matched against the purchase register, with suggested actions
GSTR-3B	Liability auto-filled from GSTR-1 cannot be edited from the July 2025 period; corrections go through GSTR-1A. Returns are barred 3 years after the due date	Validate GSTR-1 before filing; alert on returns approaching the 3-year bar
GST changes ahead	57th GST Council (Oct 2026) recommended a GSTR-3B correction mechanism from April 2027 and new reverse-charge and credit-reversal statements, pending law changes	Track; design return data so new statements can be added
E-commerce TCS and TDS	TCS 0.5% from 10 Jul 2024; TDS 0.1% (old 194-O, now section 393) (verify)	Receivable ledgers matched to GSTR-8 and Form 26AS / AIS

Area	Rule	What Progen must do
Income-tax Act 2025	In force from 1 Apr 2026: salary TDS under section 392, other TDS under section 393 (old 194C and others); Form 16 becomes Form 130 and Form 24Q becomes Form 138 (verify)	Keep old and new section references by date; contractor TDS 1% / 2%
Labour Codes	All four Codes in force from 21 Nov 2025; final central rules notified 8 May 2026	Salary-structure check for the 50% wage rule; overtime at 2× with quarterly cap; prescribed payslip and register formats; fixed-term gratuity; appointment letters
PF and ESI	EPF wage ceiling raised to ₹25,000 from 17 Sep 2026; revamped ECR; ESI 3.25% employer, 0.75% employee	Rate and ceiling tables by date; ECR and ESI return files
Professional tax and LWF	Slabs and due dates differ by state	State-wise rule tables, editable by admin
Contract labour and BOCW	Contract labour rules apply from 50 workers; BOCW cess notified at 1% of construction cost	Licence tracking; contractor registers; cess calculation and deduction
RERA	Quarterly project updates; 70% of collections in a separate account; advance capped at 10% before agreement; carpet-area pricing	Built into the real-estate sales module
Data protection (DPDP)	Rules notified Nov 2025 with an 18-month phase-in; main duties from 13 May 2027: consent per purpose, breach report within 72 hours, logs kept at least 1 year, rights requests answered within 90 days	Consent register, data minimisation for GPS and face data, retention and erasure jobs, breach workflow, rights-request portal

Integrations

Category	Integrations	Purpose
GST	E-invoice (IRP), e-way bill and GST return APIs through a licensed GST Suvidha Provider	Compliance from inside Progen
Banking and payments	Bank statement feeds and connected-banking APIs; UPI, cards and payment links (Razorpay, Cashfree, PhonePe, PayU); bulk vendor and wage payouts	Collect faster, pay in bulk, reconcile automatically
Messaging	WhatsApp Business Platform, SMS with DLT-registered templates, email	Invoices, reminders, approvals, alerts
Marketplaces and storefronts	Amazon, Flipkart, Meesho, Myntra, Ajio, Nykaa, JioMart, eBay, Shopify, WooCommerce	See E-commerce pack
Logistics	Shiprocket and other aggregators; direct courier APIs such as Delhivery and Blue Dart	Labels, tracking, NDR, COD remittance
Attendance hardware	ZKTeco, eSSL and similar biometric devices	Use existing devices
Identity and signing	Aadhaar eSign, DigiLocker, PAN, GSTIN and bank-account verification	Faster, verified onboarding of employees, vendors and buyers
Lead sources	99acres, MagicBricks, Housing.com, IndiaMART, JustDial, Meta lead ads, website forms	All enquiries in the CRM automatically
Migration	Tally, Vyapar, BUSY and Excel import; Tally export for CAs who still need it	Easy switch; no forced change for the CA
Statutory files	EPFO ECR, ESIC returns, TDS return files	Upload-ready files
Maps	Google Maps or MapmyIndia	Geo-fences, routes, distance claims

Category	Integrations	Purpose
Open platform	Public REST API, webhooks, Google and Microsoft sign-in	Customers and partners extend Progen

Non-functional requirements

These are the measurable targets every module must meet before release. The numbers are proposed starting targets for the team to confirm.

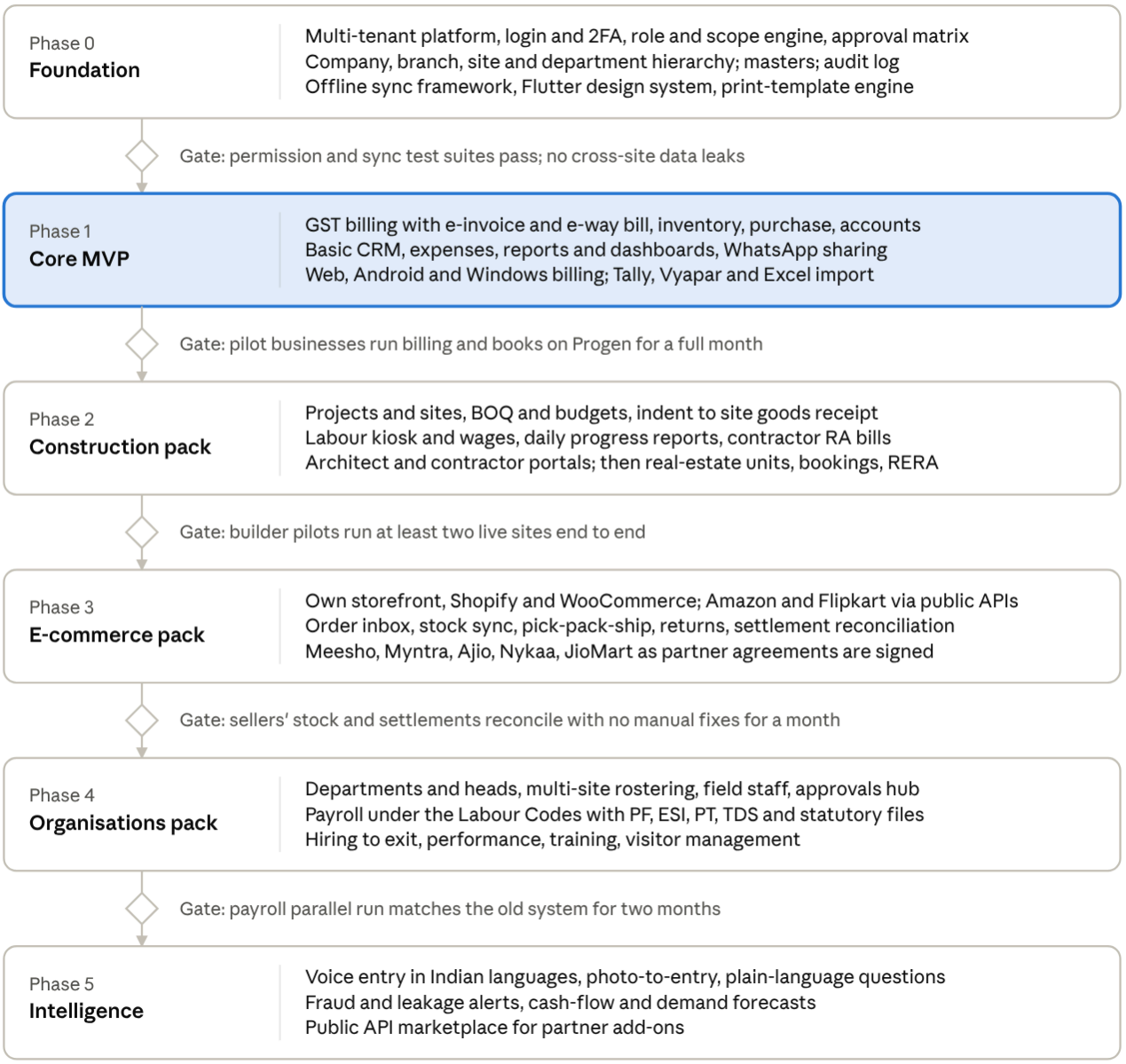
Area	Requirement	Target
Security	Encryption in transit and at rest; tenant isolation enforced in the database (row-level security), not only in code	TLS 1.2+; AES-256; zero cross-tenant reads in automated tests
Security	OWASP ASVS level 2 controls; external penetration test before each major release	No open high or critical findings at launch
Authentication	OTP / 2FA, Google and Microsoft sign-in, session timeout, device binding for field apps	2FA mandatory for admin and finance roles
Audit	Immutable log of every create, edit, cancel, approve and login, with old and new values	Retained for at least 8 years (books-of-account retention under the Companies Act)
Invoice numbering	Gapless, unique series per financial year and GST registration; cancelled numbers kept, never reused	Max 16 characters, as GST rules require
Performance	List screens and invoice save stay fast on 4G and low-end phones	List load < 2 s; invoice save < 1 s; 1 lakh-voucher report < 5 s
Scale	Per company: items, vouchers and concurrent users without slowdown	1,00,000 items; 10 lakh vouchers a year; 500 concurrent users
Availability	Hosted service uptime, backup and recovery	99.9% uptime; point-in-time recovery; RPO 15 min; RTO 4 h

Area	Requirement	Target
Offline	Billing, attendance, daily progress report, goods receipt, stock issue and picking work without internet	7 days offline; per-device number blocks so offline invoices never clash
Sync conflicts	Masters: server wins with notice. Transactions: append-only, never overwritten. Conflicts shown in a review queue	Zero silent data loss
Data residency and privacy	Data stored in India; consent for location, face and biometric data; export and erase on request (DPDP Act)	India-region hosting; consent log per employee
Data import	Excel templates for masters and opening balances; importers for Tally, Vyapar and Busy data	Opening balances and masters live in under 1 day
Data ownership	Customer can export all data at any time in open formats (Excel, CSV, JSON) and download full backups	Self-service export; no lock-in
Languages	UI and print templates in English, Hindi and major Indian languages; lakh/crore number format	Hindi + 8 regional languages by general availability
Accessibility	Large touch targets and icon-led worker kiosk; voice input for notes; web meets WCAG 2.1 AA	Usable by workers with limited literacy
Devices	Android, iOS, web browsers, Windows desktop, Bluetooth / USB thermal printers	Android 8+ on 2 GB RAM phones; iOS 15+; Windows 10+
Observability	Central logs, metrics and error tracking; integration sync log visible to the customer	Every failed marketplace or GST call visible with a retry button
Accounting integrity	Automated checks run continuously	Debits = credits for every voucher; stock ledger value = inventory account balance

Roadmap

Build the shared core first and sell it as a GST billing and accounting product. Then add each industry pack behind a pilot gate, so no pack ships before real customers have run it.

Core MVP ships first; each industry pack passes a pilot gate



Roadmap · 6 phases, 5 gates

Once the core MVP passes its gate, phases 2, 3 and 4 can run in parallel with separate teams. Construction labour wages are built in phase 2 on the core HR module; full statutory payroll follows in phase 4. Durations depend on team size, which this spec does not yet fix.

Estimated budget

The full spec costs about **₹4.3–6.7 crore** with an in-house Indian team over 18–20 months, or about ₹11–17 crore at agency rates. On top of that come about ₹35–70 lakh of one-time non-development costs and a 15% contingency. The core MVP alone (phases 0–1) is about ₹1.7–2.7 crore over 9–11 months.

Assumptions

- **Person-month** = 160 hours. In-house cost = salary × 1.3, covering PF, gratuity, office, laptops and software.
- **Salaries:** Flutter developers earn ₹11–21 LPA mid-level and ₹21–45 LPA senior (September 2026); backend, QA, design, DevOps and product roles are assumed in line. A mixed team then costs about **₹1.6–2.5 lakh per person-month**.
- **Agency rates:** Indian firms charge about $25–40 an hour mid-level and $35–55 senior, which is about **₹4.2–6.3 lakh per person-month** at ₹88 = $1. Use this column if Progen is quoted to a client or outsourced.
- **Effort** is estimated from the scope in this spec, ±25%. A two-week discovery sprint with story-level estimates should replace these figures before commitment.
- **Not included:** sales and marketing, customer support after launch, and GST on services.

Budget options

Option	Scope	Duration	Peak team	Person-months	In-house cost	Agency price
Lean MVP	Foundation + GST billing, inventory, accounts, role engine; web and Android only; Excel import only	6–7 months	7	about 45	₹72 lakh–1.1 crore	₹1.9–2.8 crore
Core MVP	Phases 0–1 as specified	9–11 months	12	about 106	₹1.7–2.7 crore	₹4.5–6.7 crore
Full spec	Phases 0–5, with phases 2–4 run by parallel teams	18–20 months	about 25	about 266	₹4.3–6.7 crore	₹11–17 crore

Recommendation: fund the core MVP first. It earns revenue as a GST billing and accounting product while the industry packs are built, and its pilot gate tests the effort estimates before the larger spend.

Cost by phase

Phase	Duration	Team	Person-months	In-house cost	Agency price
0. Foundation	3–4 months	8	28	₹45–70 lakh	₹1.2–1.8 crore
1. Core MVP	6–7 months	12	78	₹1.25–1.95 crore	₹3.3–4.9 crore
2. Construction pack	6–7 months	8	52	₹83 lakh–1.3 crore	₹2.2–3.3 crore
3. E-commerce pack	6 months	8	48	₹77 lakh–1.2 crore	₹2.0–3.0 crore
4. Organisations pack	5–6 months	7	40	₹64 lakh–1.0 crore	₹1.7–2.5 crore
5. Intelligence	4 months	5	20	₹32–50 lakh	₹84 lakh–1.3 crore

Team counts include a share of the tech lead, product manager, designer, QA and DevOps. Phase 1 is the largest because accounting, GST, inventory and three client platforms must all be production-grade before anything else can be sold.

One-time non-development costs

These total about ₹35–70 lakh over the build. Items marked "estimate" are my judgement; replace them with quotes.

Item	Cost	Basis
Security testing by a CERT-In empanelled firm, three rounds	₹8–21 lakh	₹1.5–4 lakh per round for web, API and cloud; ₹0.5–1.5 lakh per mobile platform
Domain experts on retainer: chartered accountant, construction quantity surveyor, payroll specialist	₹15–25 lakh	Estimate
Legal: terms of service, DPDP privacy notices, data-processing and marketplace partner agreements	₹3–8 lakh	Estimate
Pilot onboarding: data migration help and on-site training at each gate	₹5–10 lakh	Estimate
Test devices: low-end Android phones, iPhone, kiosk tablets, thermal printers, a biometric device	₹2–4 lakh	Estimate
Cloud for development and staging during the build	₹1–3 lakh	Estimate: ₹5,000–15,000 a month

Running costs: infrastructure and services

Running Progen as your own product costs about **₹21,000 a month during the pilot**, about **₹95,000 a month at launch** (around 200 customer companies) and about **₹4.6–5.1 lakh a month at growth** (around 2,000 companies). With a six-month pilot, that is about ₹7 lakh in the first year. Per customer company the cost falls from about ₹1,050 to about ₹240 a month, so it stays a small share of a subscription price.

Assumptions

- Pilot: up to 20 companies and 200 users. Launch: about 200 companies and 2,000 users. Growth: about 2,000 companies and 20,000 users.
- Pilot and launch run on DigitalOcean's Bangalore region: simple, and data stays in India. Growth moves to AWS Mumbai, which offers larger managed PostgreSQL servers with failover. Confirm each service is available in the region before choosing.
- Prices are on-demand, at ₹88 = $1, before 18% GST. Committing to AWS for one year cuts server and database cost by [about a third].
- Monthly messages: pilot 10,000 emails, 2,000 SMS and 5,000 WhatsApp; launch 10 times that; growth 100 times.

Hosting (per month)

Component	Pilot	Launch	Growth
App servers for API and background jobs	2 × 2 vCPU / 4 GB: ₹4,200	3 × 2 vCPU / 8 GB + load balancer: ₹17,700	6 × 4 vCPU / 16 GB: ₹44,900
PostgreSQL database	2 vCPU / 4 GB, single node: ₹5,400	6 vCPU / 16 GB with a standby for failover: ₹43,000	8 vCPU / 32 GB with standby, plus a read replica for reports: ₹1,54,000
Database storage	Included	Included	About 2 TB: ₹23,000 (approx.)
File storage for photos, bills, drawings	250 GB included: ₹440	About 1 TB: ₹2,600	About 10 TB: ₹22,000 (approx.)
Bandwidth	Included	Included	₹50,000–96,000 (approx.; lower behind a CDN)
Cache and job queue (Redis), load balancer	On the app server	₹2,600 (approx.)	₹12,000 (approx.)

Component	Pilot	Launch	Growth
Staging environment for testing releases	₹3,400	₹3,400	₹13,000 (approx.)
Hosting total	**about ₹13,500**	**about ₹69,000**	**about ₹3.2–3.7 lakh**

Prices from DigitalOcean servers, managed databases (a standby costs the same as the primary) and file storage; growth-stage servers and databases from AWS's published Mumbai price list (servers, PostgreSQL), October 2026. Items marked approx. are my estimates.

Services (per month)

Service	Pilot	Launch	Growth	Price basis
Error tracking	Free	₹2,300	₹7,000	Sentry: free, $26 or $80 a month
Email	₹90	₹900	₹8,800	Amazon SES: $0.10 per 1,000 emails
Push notifications	Free	Free	Free	Firebase Cloud Messaging
SMS for OTPs	₹500	₹4,000	₹34,000	MSG91: ₹0.25 falling to ₹0.17 per SMS with volume, plus GST
WhatsApp utility messages	₹725	₹7,300	₹72,500	₹0.145 per message through a provider
Code hosting, CI and iOS builds	₹5,000	₹10,000	₹15,000	Estimate
Domain, SSL, app store accounts	about ₹1,000	about ₹1,000	about ₹1,000	SSL free; Apple $99 a year; Google Play $25 once
GST e-invoice and e-way bill API	Quote	Quote	Quote	GSPs bill per API call; get two or three rate cards
Maps	Free tier	Usage-based	Usage-based	Check geo-fences on the phone to keep calls low
Services total	**about ₹7,300**	**about ₹25,500**	**about ₹1.4 lakh**	

How to keep it low: charge WhatsApp and SMS above a plan quota to customers, use WhatsApp instead of SMS for OTPs, serve photos through a CDN, and take one-year commitments once usage is steady. People cost is separate: a maintenance team of about 5 is ₹8–13 lakh a month. AI features in phase 5 add model API fees, best charged per plan.

Acceptance checklist

A module is "working" only when its end-to-end scenarios below pass on staging with realistic data, on web and on a low-end Android phone. Tick each one when it passes.

Roles and security

- [ ] A Site Supervisor of Site A sees no Site B record in lists, search, dashboards, reports, exports, PDFs or the API
- [ ] A hidden field (purchase rate, salary) stays hidden in screens, prints, exports, notifications and API responses
- [ ] A purchase order above the user's limit routes to the next approver; the creator cannot approve it
- [ ] Temporary access stops at its end date without manual action
- [ ] Every edit appears in the audit log with old value, new value, user, time and device

Accounting, billing and GST

- [ ] Every posted document creates balanced entries; trial balance, P&L and balance sheet tie out
- [ ] A backdated entry into a locked period is refused
- [ ] An e-invoice returns IRN and signed QR in the sandbox, and can be cancelled within the allowed window
- [ ] An e-way bill is created from the invoice and its vehicle details can be updated
- [ ] GSTR-1 and GSTR-3B data export passes the GST offline tool validation
- [ ] A bank statement import auto-matches most UPI and NEFT lines; the rest go to a review queue

Inventory and offline

- [ ] A transfer from Site A to Site B shows "in transit" until received; quantity and value reconcile on both sides
- [ ] Buying steel in tonnes and issuing in kg keeps stock and valuation correct
- [ ] 50 invoices created offline on two devices sync with no duplicate numbers and no lost entries

- [ ] Offline attendance punches sync with their original time, photo and location

Construction and builders

- [ ] BOQ → budget → indent → PO → site goods receipt → issue to activity → cost shows against the BOQ item in the project cost report
- [ ] Labour attendance on the site kiosk flows into the wage sheet and the labour contractor's payment
- [ ] A contractor running bill applies measurement, retention, advance recovery and TDS automatically
- [ ] Unit booking → payment schedule → demand letter raised when a construction stage is marked complete → receipt posted
- [ ] An architect logged in to the portal sees only drawings and RFIs of their assigned project

E-commerce

- [ ] An order placed on each connected marketplace sandbox appears in Progen and reduces stock on every channel
- [ ] A stock change in Progen updates all channels, respecting the safety buffer
- [ ] A return is received, quality-checked, restocked or written off, and a claim is raised when the item is damaged
- [ ] A marketplace settlement file matches every order's commission, fees, TCS and TDS; unmatched lines are flagged

Organisations

- [ ] An employee transferred mid-month has attendance and salary cost split between the two sites
- [ ] Leave routes to the acting head when the department head is on leave
- [ ] PF, ESI, professional tax and TDS match a statutory calculator for at least 20 sample employees
- [ ] Full and final settlement completes only after every department clears the employee

Non-functional

- [ ] Load test with 500 concurrent users meets the performance targets
- [ ] Penetration test closes with no high or critical findings
- [ ] A point-in-time restore from backup is rehearsed and documented
- [ ] A full data export opens correctly in Excel